

B.Tech. Degree VII Semester Supplementary Examination May 2024**ME 19-205-0701 REFRIGERATION AND AIR CONDITIONING***(2019 Scheme)*

Time: 3 Hours

Maximum Marks: 60

(Refrigeration data books are permitted for examination)

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of different refrigeration cycles.
 CO2: Study the vapour compression and vapour absorption refrigeration cycles- both theoretical and practical cycles, properties of refrigerants and selection criteria for refrigerants.
 CO3: Understand the different components of refrigeration systems.
 CO4: Get knowledge of psychrometry, air-conditioning systems and will be able to do the cooling load calculation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

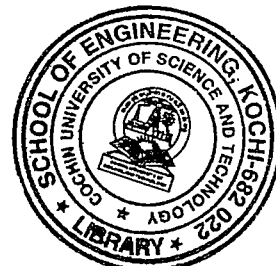
	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Explain the working of a pulse tube refrigeration system.	3	L2	1	1
(b)	Describe the boot strap air refrigeration cycle used in aircrafts.	3	L1	1	1
(c)	Discuss the effect of suction pressure and delivery pressure on the performance of vapor compression system.	3	L2	1	1
(d)	Explain the desirable physical and chemical properties of refrigerants.	3	L2	2	1
(e)	Explain the working of a centrifugal compressor.	3	L2	2	1
(f)	What are the factors affecting condenser capacity?	3	L1	3	1
(g)	What are the main differences between summer and winter air conditioning?	3	L1	4	1
(h)	Explain the comfort chart.	3	L1	4	1

PART B

(4 × 12 = 48)

- II. (a) An ice manufacturing plant works between -20°C and 40°C. The actual COP of the refrigeration plant is one third of an ideal plant working between the same temperatures. It produces 50 tonnes of ice at 0° C from water at 30° C in a day. Find the power required to drive the plant. 8 L4 1 1,2,4
- (b) Discuss the advantages of using air cycle for aircraft refrigeration. 4 L2 1 1,2
- OR**
- III. (a) An air refrigeration system operating on Bell Coleman cycle, takes in air from cold room at 268 K and compress it from 1 bar to 5.5 bar, the index of compression being 1.25. The compressed air is cooled to 300 K. The ambient temperature is 20°C. The index of compression and expansion is 1.35. Calculate: 8 L4 1 1,2,4
- (i) COP of the system
- (ii) Capacity of the plant.
- (b) Derive an expression for the efficiency of a Bell- Coleman cycle. 4 L3 1 1,2

(P.T.O.)



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		Marks	BL	CO	PO
IV.	(a) Explain the working of an aqua ammonia vapour absorption system.	4	L3	2	1,2
	(b) In a 12 tonnes R12 refrigeration plant, the condenser pressure and evaporator pressure are 12 bar and 3 bar respectively. Determine the coefficient of performance of the plant and power required to drive the plant.	8	L4	2	1,2,4
OR					
V.	(a) An NH ₃ refrigerator operates between 25°C and -15°C. The mass flow rate of the refrigerant is 0.2 kg/minute. The vapour is superheated to -10°C and the liquid is subcooled by 10°C. If the actual COP is 60% of the theoretical COP, calculate the power required to drive the system.	8	L4	2	1,2,4
	(b) Explain three fluids absorption system with a neat sketch.	4	L3	2	1,2
VI.	(a) Discuss different types of compressors used in vapour compression systems.	6	L2	3	1,2
	(b) Compare water cooled and air-cooled condensers.	6	L3	3	1,2
OR					
VII.	(a) Explain a flooded type evaporator in detail. What is the main limitation?	6	L2	3	1,2
	(b) Describe the working of thermostatic expansion valve with a neat sketch.	6	L2	3	1,2
VIII.	(a) Air at 28°C DBT and 17°C WBT flows through a cooling coil and comes out at 15°C DBT without changing moisture content. Find the initial and final relative humidity and the final wet bulb temperature.	8	L4	4	1,2,4
	(b) Explain the central and unitary systems of air conditioning. Where do we apply it?	4	L3	4	1,2
OR					
IX.	(a) Explain various heat sources to be considered for the cooling load calculation of a building for summer air conditioning.	6	L3	5	1,2
	(b) Write a short note on effective temperature and factors governing it.	6	L2	5	1,2

Blooms's Taxonomy Levels

L1 – 10%, L2 – 33.33%, L3 – 23.3%, L4 – 33.33%.
